Climate Catalyst



GREEN PUBLIC PROCUREMENT: Japan's Green Purchasing Act

About this case study

Twenty seven per cent of India's steel demand comes from public construction and infrastructure projects. With steel production expected to triple by 2050, this represents an opportunity to drive sectoral transformation while supporting economic growth.

Green Public Procurement (GPP) isn't just another policy tool, it's how governments worldwide are creating markets for low-carbon steel. In this series, we're exploring how Green Public Procurement is being used and considered by governments across the world to help drive the creation of a market for green steel products. It aims to offer examples of existing and incoming policy which could be relevant in the Indian context.

This is part of a series of case studies which we will release over the course of 2025. You can access all of the case studies at https://bit.ly/GPPinIndia

About Climate Catalyst

Climate Catalyst works to compel our political leaders to take action at speed and scale. We do this by galvanising the collective power of businesses, investors and civil society to influence and accelerate policy change. Working behind the scenes we bring together new and diverse stakeholders to identify opportunities with the greatest potential for collaboration, share ideas and learnings and deliver creative campaigns that build power and secure decisive action by governments that reduce greenhouse gas emissions. Our vision is for a just, prosperous world in which global temperature rise is limited to 1.5°C.

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Introduction

Japan's Green Purchasing Act (GPA) offers a valuable example of how public procurement can be leveraged to build new markets for sustainable products. It was one of the world's first laws requiring government ministries and agencies to give preference to environmentally friendly goods and services. Over time, the policy has helped create government-led markets for a wide variety of products, from office paper to vehicles, influencing procurement at both municipal level and in the private sector.

In January 2025, Japan extended the GPA to include steel products, one of the country's most carbon intensive industries. The decision came as part of Japan's broader Green Transformation (GX) policy package aimed at achieving net zero emissions by 2050. The amendment represents one of the first national-level efforts anywhere in the world to mainstream low-carbon steel in public procurement. There has been some criticism of the way in which the policy defines "green steel" using a "mass-balance approach", with lessons for India on the importance of matching policy aims with financial incentives. These are discussed in detail below.

For India, which is finalising its own Green Public Procurement (GPP) framework for low-carbon steel, Japan's experience demonstrates both the opportunity and the complexity of using procurement to drive industrial decarbonisation. The case underscores three essentials: transparency in standards, alignment between fiscal and procurement policy, and coordination between ministries and industry.



About the Act on Promotion of Procurement of Eco-Friendly Goods and Services

When did it come into force? The Green Purchasing Act (GPA) was introduced in 2001, and expanded to include the steel sector in January 2025.

Who does it apply to? The Green Purchasing Act (GPA) requires all national government ministries, agencies, and affiliated institutions to prioritise environmentally preferable products when making purchasing decisions.

How does it work? The GPA operates through a Basic Policy issued annually by the Ministry of the Environment, which lists categories of products and establishes green criteria for each. These criteria cover energy efficiency, recycled content, durability, emissions, and other lifecycle indicators. Local governments and private companies often mirror or voluntarily adopt these central guidelines, magnifying the policy's market influence.

In January 2025, the Cabinet approved a major amendment to the GPA's Basic Policy to include steel products in the list of designated goods. The amendment urges procuring entities to prioritise "green steel" when buying goods containing steel. Although compliance remains technically voluntary, in Japan's administrative culture, "voluntary" government guidance often leads to near-universal adherence. The amendment uses the Japan Iron and Steel Federation (JISF)'s definition of "green steel", which is based on the "mass-balanced approach" (MBA).

At this point, the amendment only applies to products made with steel. The Ministry of Land, Infrastructure, Transport and Tourism is responsible for most government procurement of steel for infrastructure projects (which account for around 13% of total steel consumption in Japan) – is considering its own green steel procurement recommendations, although it is expected to follow those set out in the Green Purchasing Act amendment

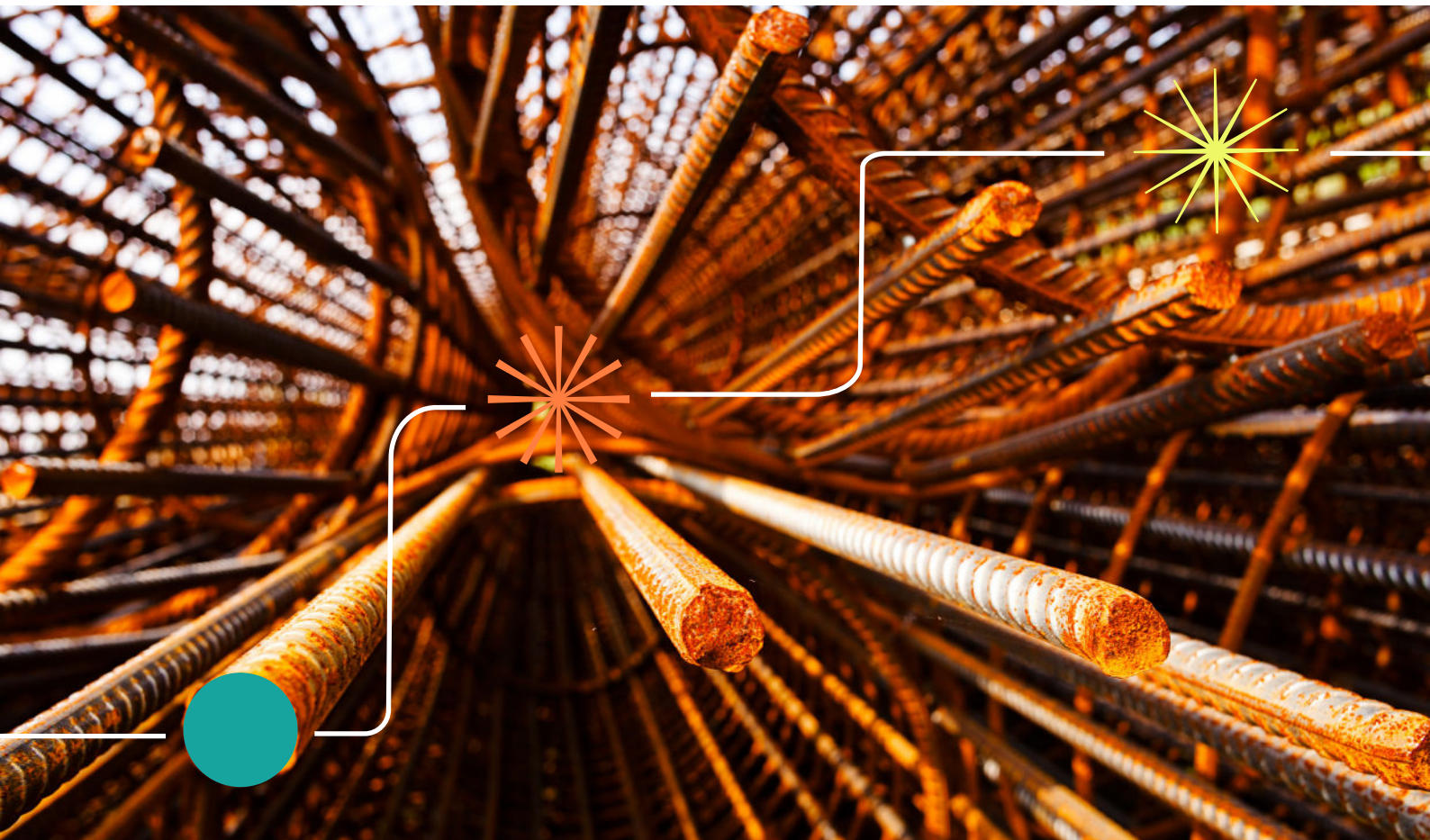


Success of GPA and its expansion to steel

Since its introduction in 2001, the GPA has become a cornerstone of Japan's sustainable procurement strategy, driving systemic change across public and private sectors. By embedding environmental criteria into purchasing decisions and fostering alignment across levels of government, the GPA not only reduces the environmental footprint of public spending but also catalyses broader market transformation toward eco-friendly goods and services.

Over the past two decades, the GPA has proven highly effective in stimulating demand for greener products, says Yuko Nishida, Senior Manager for Climate Change at Japan's Renewable Energy Institute. Nishida adds that the local government tends to also follow recommendations in the act and the guidelines prepared by the Minister of the Environment.

As it looked to extend the GPA to cover the steel sector, Japan has also launched an ambitious policy package for low carbon steel including significant state capex subsidies, opex subsidies for secondary steelmakers, lead market premiums for low carbon steel in cars, partnerships for green iron in the Middle East, and green public procurement. This "whole of economy approach" of Japan's policy package, of which GPP is a key part, has crucial lessons for India. Together, policy coordination, transparency, and the forward-looking visibility on standards and products provide companies the clarity they need to undertake capital investments. This combination also forms a package which addresses both range and depth of change necessary for decarbonisation of the steel sector.





Interactions with other policies

The inclusion of steel products within the Green Purchasing Act is designed to create a demand signal for green steel. It is also accompanied by other policies that address both the demand and supply sides of the equation.

On the supply side, these include measures within Japan's 'GX' green transformation suite of policies, introduced in 2023 to put Japan on a pathway to net zero by 2050. These include:

- Subsidies for R&D projects, under its Green Innovation Fund. Launched by the Ministry of Economy, Trade and Industry (METI) in October 2024, the fund offers subsidies to new electric-arc furnace (EAF) installations worth a total of JPY 484.4 billion (US$3.2 billion) raised through the government's Climate Transition Bond. The subsidy is targeted to current blast furnace-basic oxygen furnace (BF-BOF) steelmakers to encourage "process conversion" to new EAFs.

- Tax credits for green steel production are available to blast furnace steelmakers who replace these with EAF. The so-called 'Tax System to Promote Domestic Production in Strategic Domains' offers tax credits of JPY 20,000/tonne (US$130/tonne), to steelmakers. Similar to the subsidy programme above, these tax credits are aimed solely at new installations of EAFs, thereby encouraging conversion from carbon-intensive BF-BOF production.

Meanwhile, on the demand side, the government is also supplementing the Green Procurement Act with:

- Subsidies for clean energy vehicles using green steel. This scheme was launched in January 2025, offering JPY 50,000 (US$330) to car manufacturers for every qualifying vehicle using green steel. Analysis by Transition Asia suggests that this subsidy is likely to be sufficient to cover the likely green steel premium, even if that steel was sourced from an EAF process using direct-reduced iron produced with hydrogen, at a hydrogen price of US$5/kg.

Efficiency improvements of high carbon processes:

- The government has launched the so-called "COURSE50" and "SuperCOURSE50" programmes which focuses primarily on improving the efficiency of BF-BOFs and reducing carbon emissions.

For India, the implication is clear. With 86 per cent of India's planned BF-BOF capacity still at the pre-construction stage, a policy package that blends incentives could accelerate early transition to lower carbon steel. India has already launched the carbon market and is considering a low carbon steel subsidy scheme, a separate subsidy for secondary steel, and a GPP policy. Japan can have lessons for India on how these different tools interact with each other to make sure there is optimal use of resources for an efficiency-driven low emissions reduction pathway.

Defining “green steel”

Defining “green steel” is important. Globally, there are several standards defining what can make steel “green”. The International Energy Association (IEA)’s framework sets out definitions for “near zero emissions production” and a broader category of “low-emissions production”. The IEA definition states that “near zero emissions production” should be “physical”, meaning a firm should not be considered to have “near zero emissions” if its means of calculating reductions are reliant on “offsetting emissions from outside the supply chain or aggregation of emissions reductions credits/certificates across multiple units of production and/or supply chains” (IEA, 2024, p. 14).

The IEA’s definition includes considerations for Scope 1, Scope 2, and partial upstream Scope 3 emissions. The latter covers the production of other materials inputs in steelmaking. “Near zero emissions steel” is defined as either: (i) crude steel with no scrap use with emissions below 400kg CO2e/t, or (ii) 100% scrap-based steel with emissions lower than 50kg CO2e/t. The Climate Bonds Initiative (CBI) makes a strong distinction between primary and secondary (scrap-based) steel production, while the Steel Climate Standard established by the Global Steel Climate Council sets its own targets for carbon intensity of steel products.

National steel associations have also developed their own definitions; the German Steel Association has its own “low emissions steel standard”, while the China Iron and Steel Association has its “Methods for the assessment of China decarbonised ecological future-oriented (C2F) steel”.

When it comes to GPP, Japan has adopted a “corporate based mass-balanced approach”. This allows steelmakers to aggregate and then allocate emissions reduction from across their production units to specific steel products.

This decision has received criticism. “The reliance on the Japan Iron and Steel Federation (JISF) for the definition of green steel is seen as a significant problem by many international groups and international experts,” says Tomoko Fukunaga, climate legal fellow, at Solutions For Our Climate (SFOC). “This method favours blast furnace steel makers because it allows them to monetise small-scale decarbonisation projects and achieve a green label on their overall production without fundamentally changing their high-carbon operations. “In contrast, EAF steelmakers who already produce steel with a much lower carbon footprint do not receive similar incentives, as their emissions are already low,” she adds.

In other jurisdictions, the mass-balance approach is typically applied to an individual plant or furnace. However, the amendment to the GPA applies it at the corporate level. “The GPA allows blast-furnace companies to credit emission reductions from a single project across all of their steel output, which critics argue is a form of greenwashing that gives them an advantage over EAF steelmakers,” argues Fukunaga.

This is causing problems with private companies who have been buying lower-carbon steel from EAF producers, says Kenta Kubokawa, Japan lead at Transition Asia. “According to the ministerial order, and the JISF guidelines, even EAF steelmakers have to align themselves with the mass-balance approach,” he says. “Companies manufacturing printing machines or [steel] furniture have been delivering products made of EAF-based scrap steel to their customers for years. It is now unclear if those types of manufacturers can keep supplying products made with this steel [...] The amendment has brought huge chaos and confusion among demand-side companies”.



India's Green Steel Taxonomy, with its proposed star-rating system based on emission intensity, does not use the mass balance approach. It follows the IEA style for defining its green steel based on physical carbon emissions per tonne of finished steel rather than across production units as in Japan. "The most important lesson is to develop a clear, transparent and science-based definition of green steel," says Fukunaga. She notes that India's draft policy proposes a star rating system, "which is a good step to avoid the controversies seen in Japan". That rating system should be based on "actual physical product emissions, rather than a mass balance approach, which can be seen as greenwashing," she adds.

It is only natural that different countries will define GPP according to local circumstances. Moreover, defining "green steel" for public procurement is also dependent on meeting a variety of policy objectives beyond just carbon emissions; such as protection of the domestic steel industry or dealing with Chinese overcapacity. However, for a credible incentive to drive either efficiency improvements for existing high carbon processes or transformation investments in low-carbon steel production, the aims of the policy need to be aligned with the financial incentives.

The introduction of the GPA in Japan holds important experience from which India can learn. India has chosen to certify its green steel through the National Institute of Secondary Steel Technology (NISST) which will be responsible for verifying steel for the announced GPP programme. It is therefore vital to consider what the wider ambitions of the Indian government are when it comes to green steel production. If there is an ambition to export steel, then interoperability is an important consideration. For domestic-market only production, it is equally important to consider how the policy interacts with the myriad other policies so that the optimal outcome is achieved.

Lessons for India

Japan's experience offers a set of constructive lessons for India's emerging Green Public Procurement framework.

First, build credibility through transparency. "It is critical and indispensable that a policy has numerical standards and criteria, which are clear and evidence-based, and which are not politically influenced or which can be interpreted in different ways," says Kubokawa, describing the JISF methodology as "a black box." India's star-rating system, if based on verified emission data, could ensure that credibility from the start.

Second, align supply and demand-side instruments. Japan's success lies in how procurement standards, subsidies, and tax incentives reinforce each other. For India, aligning the Green Steel Taxonomy, National Green Hydrogen Mission, and domestic production-linked incentives will be crucial to provide both a demand signal and financial support for producers ready to transition.

The Ministry of Steel in India has announced several initiatives to support low-emission steel. A Green Steel Mission with an estimated cost of INR 15,000 crore (US$ 1.6 billion) has been proposed but not rolled out yet, with the Ministry still finalising its design. A separate INR 5,000-crore (US$ 560 million) scheme for secondary steel has also received internal approval, but is still being formulated. In addition, INR 455 crore(US$ 51 million) has been allocated for pilot projects in the steel sector under the National Green Hydrogen Mission. These efforts sit alongside other measures such as the launch of India's carbon market and the Production-Linked Incentive (PLI) scheme for speciality steel.

Third, avoid unintended bias towards resource shuffling between technologies for suboptimal emissions reductions. Japan's limited eligibility for EAF producers has been a source of tension. Given India's diverse steel sector, combining large integrated BF-BOF players, a wide base of small re-rollers, and emerging EAF producers, the incentives should be technology-neutral but performance-based, rewarding actual emission reductions rather than legacy routes. It is vital to have clarity on the aims of each scheme. For example, a programme aiming to support upgrades for BF-BOF plants to improve their efficiency should receive support which is corresponding for such improvements, while transformation projects, including EAF installations, would require much larger support.

Fourth, institutional clarity ensures longevity. Japan's long-standing GPA endures because it sits within a clear, annually updated administrative framework. India should similarly empower a single nodal agency to coordinate across ministries, set timelines, and publish verified procurement data.

Finally, procurement can normalise the "green premium." By explicitly allowing ministries and public agencies to pay a modest premium for verified green steel, India could create stable early-stage demand that signals confidence to private investors; just as Japan's clean-vehicle subsidy and procurement mandates have done.

In essence, Japan's GPA demonstrates that procurement is most powerful when integrated into a wider industrial-transition ecosystem. If India ensures transparent definitions, coherent financial alignment, and strong institutional coordination, its forthcoming GPP framework could not only decarbonise domestic steelmaking but also position India as a global leader in credible, market-shaping green procurement.

Interviewees

This case study is based on desk research and interviews with the following low-carbon procurement experts:

* Yuko Nishida, Senior Manager at the Renewable Energy Institute, Tokyo.
* Kenta Kubokawa, Japan lead at Transition Asia.
* Tomoko Fukunaga, climate legal fellow, at Solutions For Our Climate.

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